# J.A.R.V.I.S.

Setup guide: Claude-powered assistant with phone chat, phone calls, and a desktop HUD

**Do this first**

**Revoke the API key you pasted into the chat earlier.** Anything pasted into a chat is stored and should be treated as public. Go to **platform.claude.com > API Keys**, delete it, and create a new one.

From now on, keys only ever go into Jarvis's settings tool (Step 3). Never into chats, emails, or GitHub.

## How it fits together

Jarvis is a small program that runs all the time on a computer you control. It sends what you say to Claude, runs tools (web search, texting you, calling you), and answers. You reach it three ways:

| Channel | What it is | Needs |
|---|---|---|
| **Telegram** | Text Jarvis from your phone, any time | Free Telegram bot |
| **Phone call** | Jarvis rings your cell and you talk to it | Twilio number + a tunnel (ngrok) |
| **Desktop HUD** | The movie-style screen with voice, at your PC | A Windows PC with Chrome or Edge |

## Pick your path

| | Path A: Cloud server | Path B: Home PC |
|---|---|---|
| **When** | Now, from your phone | When you have a PC |
| **You get** | Telegram + phone calls, always on | Everything, including the HUD with voice and PC control |
| **Hosting cost** | About $0-4 per month | $0 (it uses your PC) |
| **Hardest part** | Creating the server from a phone browser | Nothing; double-click an installer |

**Recommendation**

Start with **Path A, Telegram only**. It's the cheapest way to have a working Jarvis today. Add phone calls once you're happy, and move to Path B when you have a PC. Your settings carry over.

### What it costs (approximate, check current prices)

| Item | Price | Notes |
|---|---|---|
| **Claude API** | Pay per use | Sonnet 5.5: $2 / $10 per million input / output tokens. Opus 5.5: $4 / $20. A short back-and-forth on Sonnet is typically well under a cent. Prepaid credit; you set a monthly limit. |
| **Telegram** | Free | |
| **Cloud server** | $0-4 / month | Google Cloud's e2-micro VM is free, but Google bills its public IP (about $0.005/hour, roughly $3.65/month). Oracle Cloud's Always Free tier is $0 but harder to set up. |
| **Twilio (calls)** | $1.15 / month + per minute | Local US number $1.15/month; outbound US calls about $0.013-0.014/minute, plus speech-recognition fees. New accounts get trial credit. |
| **ngrok (tunnel)** | Free | Free plan includes one fixed web address, which is all Jarvis needs. |

**Realistic budget for light personal use: about $5-10 per month** on Sonnet with occasional calls. This is an estimate. Your spend limit in Step 1 guarantees it can't run away.

# Step 1: Claude API key

This is how Jarvis pays for Claude. It is separate from any claude.ai subscription.

1. Open **platform.claude.com** and sign in (or sign up).
2. Go to **Billing** and add a small amount of prepaid credit (for example $5-10).
3. Go to **Limits** and set a **monthly spend limit** you're comfortable with. This is your safety net.
4. Go to **API Keys > Create Key**, name it **jarvis**, and copy it. It starts with `sk-ant-api`. Keep it in a locked notes app or password manager until Step 3.

### Choosing a model (you'll pick it in Step 3)

| Model | Best for | Cost |
|---|---|---|
| **Sonnet 5.5** | Everyday assistant use. **Recommended for low cost.** | About half of Opus |
| **Opus 5.5** | The smartest answers and trickier multi-step tasks | Highest |
| **Haiku 5.5** | The cheapest option; no web search | Lowest |

# Step 2: Telegram bot (5 minutes, on your phone)

1. Install **Telegram** and create an account.
2. Search for **@BotFather** (it has a blue check mark) and open the chat.
3. Send `/newbot`. Give it a name (e.g. *Jarvis*) and a username that ends in *bot* (e.g. *tony_jarvis_bot*).
4. BotFather replies with a **token** like `123456789:AAF...`. Copy it and keep it with your API key.

You'll get your **chat id** later: the first time you message your bot, Jarvis replies with it.

# Path A: Run Jarvis on a cloud server

These steps use Google Cloud because its **SSH** button opens a terminal right in your browser, which works from a phone. Turn on **Desktop site** in your phone browser's menu: the console is much easier to use that way.

## A1. Create the account and a budget alert

1. Go to **console.cloud.google.com** and sign up. A card is required for verification.
2. Open **Billing > Budgets & alerts > Create budget**. Set it to **$5** so you're emailed if anything costs more than expected.

## A2. Create the server

1. Go to **Compute Engine > VM instances** (click **Enable** if asked) and press **Create instance**.
2. **Name:** jarvis. **Region:** us-central1, us-west1 or us-east1 (only these are in the free tier).
3. **Machine type: e2-micro**.
4. **Boot disk > Change:** Ubuntu 24.04 LTS, disk type **Standard persistent disk**, size 30 GB. The default disk type is *not* free, so change it.
5. Leave everything else as is and press **Create**. Wait for the green check mark.

## A3. Install Jarvis

Press the **SSH** button next to your VM. A terminal opens. Type (or paste) these lines one at a time:

```
git clone https://github.com/cmdevsarnia/jarvis.git
cd jarvis
bash deploy/setup-server.sh
```

**Private repository?**

If the repository is private, GitHub asks for a username and password. Use your GitHub username, and for the password a **personal access token** (GitHub > Settings > Developer settings > Personal access tokens > Fine-grained, read-only access to this one repository).

### A4. Enter your settings

```
.venv/bin/python -m jarvis.configure
```

It asks one question at a time: paste your Claude key, pick model **1** (Sonnet), enter what Jarvis should call you, and paste your Telegram token. Leave the chat id blank for now, and answer **N** to phone calls. Your keys are hidden while you type and stored readable only by you. Then start Jarvis:

```
sudo systemctl restart jarvis
```

### A5. Connect Telegram

1. In Telegram, open your bot and send *hello*. It replies with **Your chat id is ...**
2. Back in the SSH window run `.venv/bin/python -m jarvis.configure` again. Press Enter to keep each existing value and paste the chat id when asked.
3. Run `sudo systemctl restart jarvis`.
4. Message the bot: *Jarvis, what's the server's CPU load?* You should get a reply in a few seconds.

Jarvis now starts automatically whenever the server boots. You can close the SSH window. Useful commands: `systemctl status jarvis` (is it running?), `journalctl -u jarvis -n 50` (last 50 log lines).

### Free alternative: Oracle Cloud

Oracle's **Always Free** tier includes a VM *and* its public IP at $0. Create an Ubuntu instance (shape VM.Standard.E2.1.Micro or Ampere A1), connect with its **Cloud Shell** or the **Termius** app using the SSH key Oracle gives you, then follow A3-A5 exactly as written.

## Path B: Run Jarvis on a Windows PC

1. Install **Python 3.12 or newer** from python.org. On the first installer screen, tick **Add python.exe to PATH**.
2. Get the code: on the GitHub page for the repository, press **Code > Download ZIP** and unzip it to e.g. `Documents\Jarvis`.
3. Double-click **start-jarvis.bat**. The first run installs everything (a few minutes) and then asks your settings, the same questions as Path A.
4. Jarvis opens the HUD in your browser. Use **Microsoft Edge** for the best voice, and allow the microphone.
5. Hold **Space** (or click **MIC**) and talk. Tick **WAKE WORD** to just say *"Jarvis, ..."* hands-free.
6. Connect Telegram as in A5. To change settings later, open a terminal in the folder and run `.venv\Scripts\python -m jarvis.configure`.
7. Optional: edit `config\apps.json` to list apps Jarvis may open for you.

> **Running both?**
> Only one Jarvis can use a Telegram bot at a time. If you move from the cloud server to your PC, stop the server's copy first (`sudo systemctl disable --now jarvis`) or create a second bot.

Jarvis only runs while the window is open and the PC is awake. To start it automatically, add **start-jarvis.bat** to Windows Task Scheduler with the trigger *At log on*.

# Add-on: Phone calls

Jarvis can ring your cell and hold a spoken conversation. Do this after Telegram works.

## C1. Twilio number

1. Sign up at **twilio.com**. During signup, verify your own cell number.
2. In the Console, **Phone Numbers > Buy a number**: pick one with **Voice** capability (about $1.15/month).
3. From the Console home page, copy your **Account SID** and **Auth Token**.

On a trial account, Twilio can only call your verified number and plays a short trial message first. Upgrading removes that.

## C2. Tunnel (so Twilio can reach Jarvis)

1. Sign up free at **ngrok.com**. In the dashboard, copy your **authtoken**, then under **Domains** claim your free static domain (looks like `something.ngrok-free.app`).
2. **Cloud server:** in the SSH window run the line below and paste the token and domain when asked. It keeps the tunnel running permanently.

```
bash deploy/setup-ngrok.sh
```

**Windows PC:** download ngrok for Windows, then in a second window run the command shown in your ngrok dashboard. It looks like `ngrok http --url=YOUR-DOMAIN 8000`. Keep that window open.

## C3. Tell Jarvis and test

1. Run the settings tool again and answer **y** to phone calls. Enter the SID, token, your Twilio number and your cell (both with + and country code, e.g. `+15551234567`), and the public URL `https://YOUR-DOMAIN`.
2. Restart Jarvis.
3. Text your bot: *call me*. Your phone rings. Talk normally. Say *goodbye* to hang up.

Calling *into* Jarvis is off by default, because caller ID can be faked. Jarvis only ever calls your number.

# Troubleshooting

| Symptom | Fix |
|---|---|
| **Bot never replies** | Check `systemctl status jarvis` and the logs. Make sure the token is right and only one copy of Jarvis is running (logs show *409 Conflict* otherwise). |
| **Bot replies with your chat id every time** | TELEGRAM_OWNER_CHAT_ID isn't set yet. Run the settings tool and restart. |
| ***My API key was rejected*** | The Claude key is wrong or revoked. Make a new one and run the settings tool. |
| ***I'm being rate limited*****, or errors about credit** | Add credit or raise the limit at platform.claude.com. |
| ***Phone calls are not configured*** | One of the Twilio settings or PUBLIC_BASE_URL is blank. |
| **Call connects, then *an application error has occurred*** | Twilio can't reach Jarvis. Check the tunnel is running and PUBLIC_BASE_URL exactly matches the ngrok domain (https, no trailing slash). |
| **Phone doesn't ring on a trial account** | Your cell must be a verified number in Twilio. |
| **HUD says *Access token missing*** | Open the exact HUD link printed in the Jarvis window (it includes a token). |
| **Microphone does nothing** | Use Chrome or Edge, and allow microphone access when the browser asks. |

# Safety checklist

- The key you pasted in chat is revoked, and a spend limit is set on platform.claude.com.
- Keys exist only in Jarvis's settings (.env file), which is never uploaded to GitHub.
- TELEGRAM_OWNER_CHAT_ID is set, so Jarvis ignores everyone else.
- Inbound calls stay off unless you understand the caller-ID risk.
- Jarvis can't run arbitrary commands. Each tool is deliberately narrow, so a stolen phone can't take over your PC.
- Calling other people or businesses for you is not built in yet. When it is, it should say it's an AI and only call numbers you approve. AI-voice calls are regulated in the US.

## Everyday commands (cloud server)

| What | Command |
|---|---|
| **Is Jarvis running?** | `systemctl status jarvis` |
| **Recent logs** | `journalctl -u jarvis -n 50` |
| **Change settings** | `cd ~/jarvis && .venv/bin/python -m jarvis.configure` |
| **Restart** | `sudo systemctl restart jarvis` |
| **Update to the latest code** | `cd ~/jarvis && git pull && sudo systemctl restart jarvis` |
| **Clear the conversation** | Send `/reset` to the bot |